SQL Quick Reference

Database Management Systems - One-page cheat sheet

Query skeleton

```
SELECT [DISTINCT] columns | aggregates
FROM table [JOIN other ON condition]
[WHERE row_condition]
[GROUP BY columns [HAVING group_condition]]
[ORDER BY column [ASC|DESC]] [LIMIT n];
```

Aggregates

- COUNT(*), COUNT(col) ignores NULL, SUM, AVG, MIN, MAX.
- Use COUNT(DISTINCT col) to count unique values.

Data definition and manipulation

```
CREATE TABLE t (id INT PRIMARY KEY, name TEXT NOT NULL);
ALTER TABLE t ADD COLUMN age INT;
INSERT INTO t (id, name) VALUES (1, 'Asha');
UPDATE t SET age = 21 WHERE id = 1;
DELETE FROM t WHERE id = 1;
DROP TABLE t;
```

Constraints

- PRIMARY KEY, UNIQUE, NOT NULL, CHECK (age >= 0), FOREIGN KEY ... REFERENCES.
- ON DELETE CASCADE removes child rows automatically.

NULL handling

- Use IS NULL / IS NOT NULL - never = NULL.
- COALESCE(a, b) returns the first non-NULL value.